Compact Suffix Rules for Stemming in Morphologically Rich Languages

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Abstract

Stemming reduces inflected words to a common form and remains a cheap and effective step in retrieval pipelines. We present a compact set of suffix rules that can be learned from a few thousand word pairs. On five morphologically rich languages the rules recover 93.8% of the accuracy of a full morphological analyser while using less than one percent of its memory. We release the rule sets and the evaluation scripts.

1 Introduction

Morphologically rich languages encode grammatical information in long chains of suffixes. A single Finnish noun can appear in more than two thousand surface forms. Search engines that index only surface forms therefore miss many relevant documents.

Full morphological analysers solve this problem but need large lexicons and careful maintenance. Light stemmers need little data but make frequent errors on irregular words. We ask how far a small set of learned rules can close the gap between the two.

Our contributions are as follows:

- a compact rule format that fits in a few kilobytes;
- a learning procedure that needs only lemma and form pairs;
- an evaluation on five languages with released test sets.

2 Method

2.1 Rule format

A rule removes a suffix and optionally appends a replacement. Rules are tried from the longest suffix to the shortest, and the first rule whose condition holds is applied. Two example rules are shown below.

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rule: -ations -> -ate
rule: -ing -> "" if stem >= 3
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The condition on stem length prevents short words such as sing from being reduced to a meaningless fragment.

2.2 Learning the rules

We collect all pairs of an inflected form and its lemma. For each pair the longest common prefix is found, and the remaining suffixes define a candidate rule. A candidate is kept if it is correct for at least 95% of the pairs in which it applies. The final list is sorted by suffix length.

2.3 Evaluation

We measure the F1 score, the harmonic mean of precision P and recall R:

$$F1 = 2PR / (P + R)$$

Scores are averaged over ten random splits of the test pairs.

2.4 Data

The word pairs come from public treebanks. For each language we use 4,000 pairs for learning and 1,000 pairs for testing. Pairs that appear in both sets are removed from the test set.

3 Results

Table 1 reports accuracy on the held-out pairs. The learned rules outperform the light stemmer on every language and come within a few points of the full analyser.

Table 1: Accuracy (%) of three stemmers on held-out word pairs.

Language	Light	Rules	Full
Finnish	71.2	88.9	96.1
Turkish	74.8	90.3	95.7
Hungarian	69.5	87.6	94.9
Czech	77.3	91.8	96.4
Polish	76.1	91.0	96.0
Mean	73.8	89.9	95.8

The largest gains appear for Finnish and Hungarian, where long suffix chains are common. Errors that remain are mostly irregular stems, which no suffix rule can repair.

The rule sets occupy between 18 and 41 kilobytes, against six megabytes or more for the lexicon of the full analyser. Table 2 gives the exact numbers.

Table 2: Number of rules and size of each rule set.

Language	Rules	Size (KB)
Finnish	412	41
Turkish	355	36
Hungarian	389	38
Czech	201	21
Polish	174	18

Finnish needs the most rules because of its many cases and possessive suffixes.

3.1 Error analysis

We inspected one hundred errors for each language and sorted them into three groups:

1. irregular stems, such as vowel changes inside the word;
2. homographs whose lemma depends on context;
3. foreign words and proper names.

Irregular stems account for the largest group in every language. Homographs are rare in Finnish but frequent in Czech, where many endings are shared by different cases. Foreign words and proper names are best handled by a short exception list, which we leave to future work.

4 Conclusion

We have shown that a few hundred suffix rules recover most of the benefit of a full morphological analyser. Future work will extend the rule format to prefixes and infixes, and will test the rules in a complete retrieval system.

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References

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